

02_Requirement Analysis

FitBuddy – AI Fitness Plan Generator using Gemini Models

1. Functional Requirements

- The application shall allow users to enter the required personal details.
- The application shall allow users to select a fitness goal such as weight loss, muscle gain, or general wellness.
- The application shall allow users to select their preferred workout intensity.
- The application shall generate a personalized 7-day workout plan using Gemini AI.
- The application shall provide nutrition and recovery tips.
- The application shall allow users to provide feedback on the generated plan.
- The application shall use the feedback to refine or update the fitness plan.
- The application shall store relevant application data using SQLite.

2. Non-Functional Requirements

- Usability: The interface should be simple and easy for users to understand.
- Performance: The application should generate and display the requested plan within a reasonable time.
- Reliability: The application should handle valid user inputs and provide appropriate responses.
- Maintainability: The project should be organized into separate modules for easier maintenance.
- Security: Sensitive configuration information such as API keys should not be exposed in the source code.
- Compatibility: The application should run in a suitable Python environment with the required dependencies installed.

3. Technology Requirements

Category	Requirement
Programming / Backend	Python, FastAPI
Frontend	HTML, CSS, JavaScript, Jinja2 templates
Database	SQLite with SQLAlchemy
AI Service	Google Gemini models using the Google GenAI SDK
Data Validation	Pydantic / Pydantic Settings
Testing	Pytest and HTTPX
Development Tool	Visual Studio Code
Version Control	Git and GitHub

4. Project Dependencies

The project requires the Python packages and dependencies specified in the project's requirements.txt file. These dependencies support the FastAPI application, templates, database, configuration, Gemini AI integration, and testing.

Package / Dependency	Purpose
fastapi	Web framework and API development
uvicorn	Running the FastAPI application
jinja2	Server-side HTML templates
sqlalchemy	Database access and ORM
pydantic	Data validation and schemas
pydantic-settings	Application configuration
python-dotenv	Environment variable support
google-generativeai	Google Gemini AI integration
pytest	Application testing
httpx	HTTP client support for testing

5. Requirement Summary

FitBuddy requires a web-based FastAPI application that accepts user fitness information, uses Google Gemini models to generate a personalized 7-day plan, provides nutrition and recovery guidance, supports feedback-based refinement, and stores relevant data using SQLite. The application should be usable, maintainable, reliable, and secure.

Note: This document covers the Requirement Analysis phase. The actual project source code belongs to the Project Development phase.